Stan (Marginal Likelihood) vs blavaan: Timing Benchmark

Bayesian Polychoric Correlations in the ACE Twin Model

Overview

Two Bayesian approaches for estimating polychoric correlations in the ACE twin model are compared:

Feature	Stan (marginal likelihood)	blavaan (data augmentation)
Likelihood	Multinomial on $K(K+1)/2$ reduced cell counts	Multivariate normal on N augmented latent variables
Computational scaling	$O(K^2)$ — independent of N	$O(N)$ — linear in sample size
Implementation	`ace_ordinal_symmetry_for_compare.stan`	`bcfa()` from the *blavaan* package

Both methods use identical MCMC settings (3 chains, 500 warmup + 1000 post-warmup iterations) and equivalent priors:

- Polychoric correlations: Uniform$(-1, 1)$ — Stan uses `scaled_rho ~ beta(1, 1)` with a linear rescaling; blavaan uses `lkj_corr(1)`, which is also uniform for a bivariate case.
- Thresholds: `normal(0, 1.5)` in both.

Setup

```
library(cmdstanr)
library(lavaan)
library(blavaan)
library(ggplot2)
library(dplyr)
library(tidyr)
library(here)
# Run 3 chains in parallel
options(mc.cores = 3)
options(future.globals.maxSize = 2 * 1024^3) # 2 GB
```

Compile Stan Model

```
ace_cmp <- cmdstan_model(here("stan", "ace_ordinal_symmetry_for_compare.stan"))
```

Data

BMI category data (normal / overweight / obese) for MZ and DZ twin pairs, based on the summary frequencies from Jøreskog (2021).

```
n_mz <- c(325, 60, 7, 78, 150, 40, 8, 38, 88)
n_dz <- c(202, 97, 19, 96, 112, 64, 46, 56, 66)
dat_mz <- data.frame(
    twin1 = rep(rep(1:3, 3), n_mz),
    twin2 = rep(rep(1:3, each = 3), n_mz)
)
dat_dz <- data.frame(
    twin1 = rep(rep(1:3, 3), n_dz),
    twin2 = rep(rep(1:3, each = 3), n_dz)
)
dat_mz[] <- lapply(dat_mz, ordered, labels = c("normal", "overweight", "obese"))
dat_dz[] <- lapply(dat_dz, ordered, labels = c("normal", "overweight", "obese"))
dat_full <- rbind(
    data.frame(zygosity = "MZ", dat_mz),
    data.frame(zygosity = "DZ", dat_dz)
)

# Fixed-seed subsamples
set.seed(1234)
dat_100 <- dat_full[sample.int(nrow(dat_full), 100), ]
dat_500 <- dat_full[sample.int(nrow(dat_full), 500), ]

cat(
    "N full =",
    nrow(dat_full),
    "  (MZ:",
    sum(dat_full$zygosity == "MZ"),
    "| DZ:",
    sum(dat_full$zygosity == "DZ"),
    ")\n"
)
```

```
N full = 1552   (MZ: 794 | DZ: 758 )
```

blavaan Model Syntax

```
# Equal thresholds for both twins and across zygosity groups (same labels)
ace_blavaan_mod <- "
```

```
    twin1 ~~ twin2
    twin1 | th1 * t1 + th2 * t2
    twin2 | th1 * t1 + th2 * t2
"
```

Helper Functions

```
# Reduce a contingency table to lower-triangular vector (exploiting symmetry)
to_ltri <- function(tab_x) {
    symm_x <- tab_x + t(tab_x) - diag(diag(tab_x))
    symm_x[lower.tri(symm_x, diag = TRUE)]
}

# Run Stan marginal likelihood approach; return timing + estimates
run_stan <- function(dat, seed = 42) {
    tab_mz <- with(dat[dat$zygosity == "MZ", ], table(twin1, twin2))
    tab_dz <- with(dat[dat$zygosity == "DZ", ], table(twin1, twin2))
    fit <- ace_cmp$sample(
        list(
            k = 3L,
            ltri_ns1 = to_ltri(tab_mz), # MZ → rho1
            ltri_ns2 = to_ltri(tab_dz) # DZ → rho2
        ),
        chains = 3,
        iter_warmup = 500,
        iter_sampling = 1000,
        seed = seed,
        refresh = 0
    )
    summ <- fit$summary(c("rho1", "rho2"))
    list(
        elapsed = fit$time()$total,
        rhat_max = max(summ$rhat, na.rm = TRUE),
        rho = tibble(
            group = c("MZ", "DZ"),
            mean = summ$mean,
            sd = summ$sd,
            ci_lower = summ$q5,
            ci_upper = summ$q95
        )
    )
}
```

```
# Run blavaan data augmentation approach; return timing + estimates
run_blavaan <- function(dat, seed = 42) {
    fit <- bcfa(
        ace_blavaan_mod,
        data = dat,
        ordered = c("twin1", "twin2"),
        group = "zygosity",
        n.chains = 3,
        burnin = 500,
        sample = 1000,
        seed = seed
    )
    pe <- parameterEstimates(fit)
    summ <- summary(fit)
    cor_rows <- subset(pe, op == "~~" & lhs == "twin1" & rhs == "twin2")
    rows_id <- as.integer(rownames(cor_rows))
    # Groups are ordered alphabetically by blavaan/lavaan: DZ = 1, MZ = 2
    group_labs <- fit@Data@group.label
    cor_rows$group_label <- group_labs[cor_rows$group]
    # Max Rhat across correlation parameters (from blavaan's psrf diagnostic)
    psrf <- blavaan::blavInspect(fit, "psrf")
    list(
        elapsed = fit@timing$Estimate,
        rhat_max = max(psrf, na.rm = TRUE),
        rho = tibble(
            group = cor_rows$group_label,
            mean = cor_rows$est,
            sd = as.numeric(summ[rows_id, " Post.SD"]),
            ci_lower = as.numeric(summ[rows_id, "pi.lower"]),
            ci_upper = as.numeric(summ[rows_id, "pi.upper"])
        )
    )
}
```

Benchmarks

Results are cached to RDS files to avoid re-running.

N = 100

```
cache_100 <- here("notebooks", "bench_blavaan_100.rds")
if (!file.exists(cache_100)) {
    res_100 <- list(
        stan = run_stan(dat_100, seed = 42),
```

```
        blavaan = run_blavaan(dat_100, seed = 42)
    )
    saveRDS(res_100, cache_100)
} else {
    res_100 <- readRDS(cache_100)
}
cat(sprintf(
    "Stan: %.1f s  (max Rhat = %.3f)\nblavaan: %.1f s  (max Rhat = %.3f)\n",
    res_100$stan$elapsed,
    res_100$stan$rhat_max,
    res_100$blavaan$elapsed,
    res_100$blavaan$rhat_max
))
```

```
Stan: 0.5 s  (max Rhat = 1.001)
blavaan: 15.5 s  (max Rhat = 1.001)
```

N = 500

```
cache_500 <- here("notebooks", "bench_blavaan_500.rds")
if (!file.exists(cache_500)) {
    res_500 <- list(
        stan = run_stan(dat_500, seed = 42),
        blavaan = run_blavaan(dat_500, seed = 42)
    )
    saveRDS(res_500, cache_500)
} else {
    res_500 <- readRDS(cache_500)
}
cat(sprintf(
    "Stan: %.1f s  (max Rhat = %.3f)\nblavaan: %.1f s  (max Rhat = %.3f)\n",
    res_500$stan$elapsed,
    res_500$stan$rhat_max,
    res_500$blavaan$elapsed,
    res_500$blavaan$rhat_max
))
```

```
Stan: 0.7 s  (max Rhat = 1.002)
blavaan: 86.8 s  (max Rhat = 1.005)
```

Full Data (N = 1552)

```
cache_full <- here("notebooks", "bench_blavaan_full.rds")
if (!file.exists(cache_full)) {
    res_full <- list(
        stan = run_stan(dat_full, seed = 42),
        blavaan = run_blavaan(dat_full, seed = 42)
    )
    saveRDS(res_full, cache_full)
} else {
    res_full <- readRDS(cache_full)
}
cat(sprintf(
    "Stan: %.1f s  (max Rhat = %.3f)\nblavaan: %.1f s  (max Rhat = %.3f)\n",
    res_full$stan$elapsed,
    res_full$stan$rhat_max,
    res_full$blavaan$elapsed,
    res_full$blavaan$rhat_max
))
```

```
Stan: 1.0 s  (max Rhat = 1.002)
blavaan: 568.6 s  (max Rhat = 1.000)
```

Results

Timing

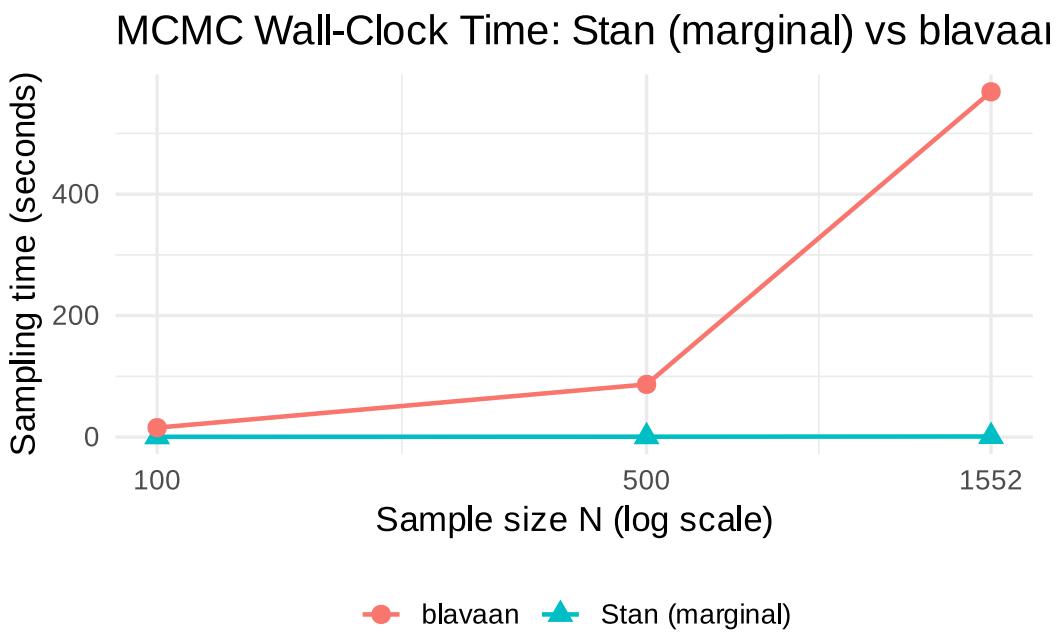
```
timing_df <- tibble(
    N = rep(c(100, 500, nrow(dat_full)), each = 2),
    Method = rep(c("Stan (marginal)", "blavaan"), 3),
    Time_s = c(
        res_100$stan$elapsed,
        res_100$blavaan$elapsed,
        res_500$stan$elapsed,
        res_500$blavaan$elapsed,
        res_full$stan$elapsed,
        res_full$blavaan$elapsed
    )
)

timing_df |>
    pivot_wider(names_from = Method, values_from = Time_s) |>
    mutate(
```

```
        `Speedup (×)` = round(`blavaan` / `Stan (marginal)`, 1),
        across(where(is.double), \(x) round(x, 1))
    ) |>
    knitr::kable(caption = "Wall-clock sampling time (seconds) by sample size")
```

N	Stan (marginal)	blavaan	Speedup (×)
100	0.5	15.5	32.5
500	0.7	86.8	133.4
1552	1.0	568.6	560.2

```
ggplot(timing_df, aes(x = N, y = Time_s, color = Method, shape = Method)) +
    geom_line(linewidth = 0.8) +
    geom_point(size = 3) +
    scale_x_log10(
        breaks = c(100, 500, nrow(dat_full)),
        labels = c("100", "500", as.character(nrow(dat_full)))
    ) +
    labs(
        x = "Sample size N (log scale)",
        y = "Sampling time (seconds)",
        title = "MCMC Wall-Clock Time: Stan (marginal) vs blavaan",
        color = NULL,
        shape = NULL
    ) +
    theme_minimal(base_size = 13) +
    theme(legend.position = "bottom")
```



Polychoric Correlation Estimates

Posterior mean and 90% credible interval for the MZ and DZ polychoric correlations (ρ_{MZ} and ρ_{DZ}) across the two methods and three sample sizes.

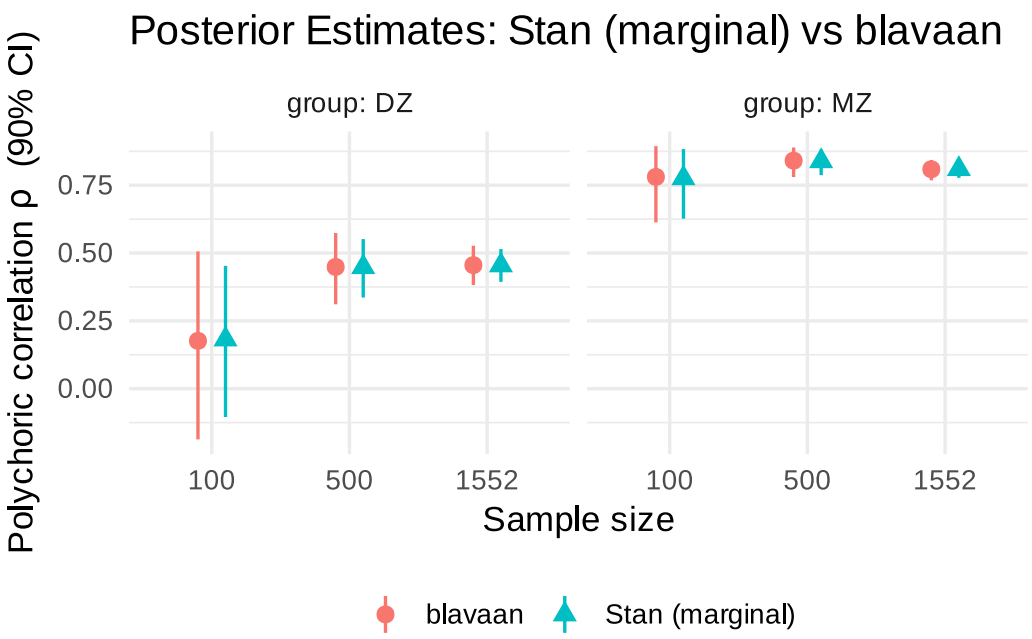
```
ests_df <- bind_rows(
    mutate(res_100$stan$rho, N = 100, Method = "Stan (marginal)"),
    mutate(res_100$blavaan$rho, N = 100, Method = "blavaan"),
    mutate(res_500$stan$rho, N = 500, Method = "Stan (marginal)"),
    mutate(res_500$blavaan$rho, N = 500, Method = "blavaan"),
    mutate(res_full$stan$rho, N = nrow(dat_full), Method = "Stan (marginal)"),
    mutate(res_full$blavaan$rho, N = nrow(dat_full), Method = "blavaan")
)

ests_df |>
    arrange(N, group, Method) |>
    mutate(across(where(is.numeric), \(x) round(x, 3))) |>
    knitr::kable(
        col.names = c("Group", "Mean", "SD", "CI 5%", "CI 95%", "N", "Method"),
        caption = "Posterior summaries of polychoric correlations (90% CI)"
    )
```

Group	Mean	SD	CI 5%	CI 95%	N	Method
DZ	0.182	0.170	−0.104	0.453	100	Stan (marginal)

Group	Mean	SD	CI 5%	CI 95%	N	Method
DZ	0.176	0.179	−0.187	0.506	100	blavaan
MZ	0.776	0.080	0.627	0.883	100	Stan (marginal)
MZ	0.781	0.076	0.613	0.894	100	blavaan
DZ	0.448	0.066	0.336	0.551	500	Stan (marginal)
DZ	0.449	0.068	0.311	0.574	500	blavaan
MZ	0.839	0.029	0.787	0.881	500	Stan (marginal)
MZ	0.841	0.028	0.780	0.889	500	blavaan
DZ	0.455	0.037	0.394	0.515	1552	Stan (marginal)
DZ	0.455	0.037	0.382	0.527	1552	blavaan
MZ	0.810	0.019	0.777	0.838	1552	Stan (marginal)
MZ	0.809	0.019	0.768	0.843	1552	blavaan

```
ggplot(
    ests_df,
    aes(
        x = factor(N),
        y = mean,
        ymin = ci_lower,
        ymax = ci_upper,
        color = Method,
        shape = Method
    )
) +
    geom_pointrange(position = position_dodge(width = 0.4)) +
    facet_wrap(~group, labeller = label_both) +
    labs(
        x = "Sample size",
        y = expression("Polychoric correlation " * rho ~ " (90% CI)"),
        title = "Posterior Estimates: Stan (marginal) vs blavaan",
        color = NULL,
        shape = NULL
    ) +
    theme_minimal(base_size = 13) +
    theme(legend.position = "bottom")
```


Posterior Estimates: Stan (marginal) vs blavaan
group: DZ
group: MZ
Polychoric correlation ρ (90% CI)
0.75
0.50
0.25
0.00
100
500
1552
100
500
1552
Sample size
blavaan
Stan (marginal)